

HDOC Day-16 Report

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Menu-Driven C Program Using Switch-Case

Digit-based operations on a positive integer using separate user-defined void functions for each menu option.

Problem Statement

Write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. Use one separate user-defined function for each menu option. Each function has a **void** return type, reads its required input, processes it, and displays the result.

Required Operations

- Check whether the digits are in increasing order from left to right.
- Check whether the digits are in decreasing order from left to right.
- Find the largest digit and the smallest digit.
- Find the second largest digit and second smallest digit in the number.

Algorithm

1. Start the program.
2. Display four menu options.
3. Read the user's choice.
4. Use switch-case to call the corresponding void function.
5. For increasing/decreasing order, compare adjacent digits while processing the number.
6. For largest/smallest, inspect every digit and update the current largest and smallest values.
7. For second largest/second smallest, maintain both first and second distinct extreme values.
8. Display the result and terminate the program.

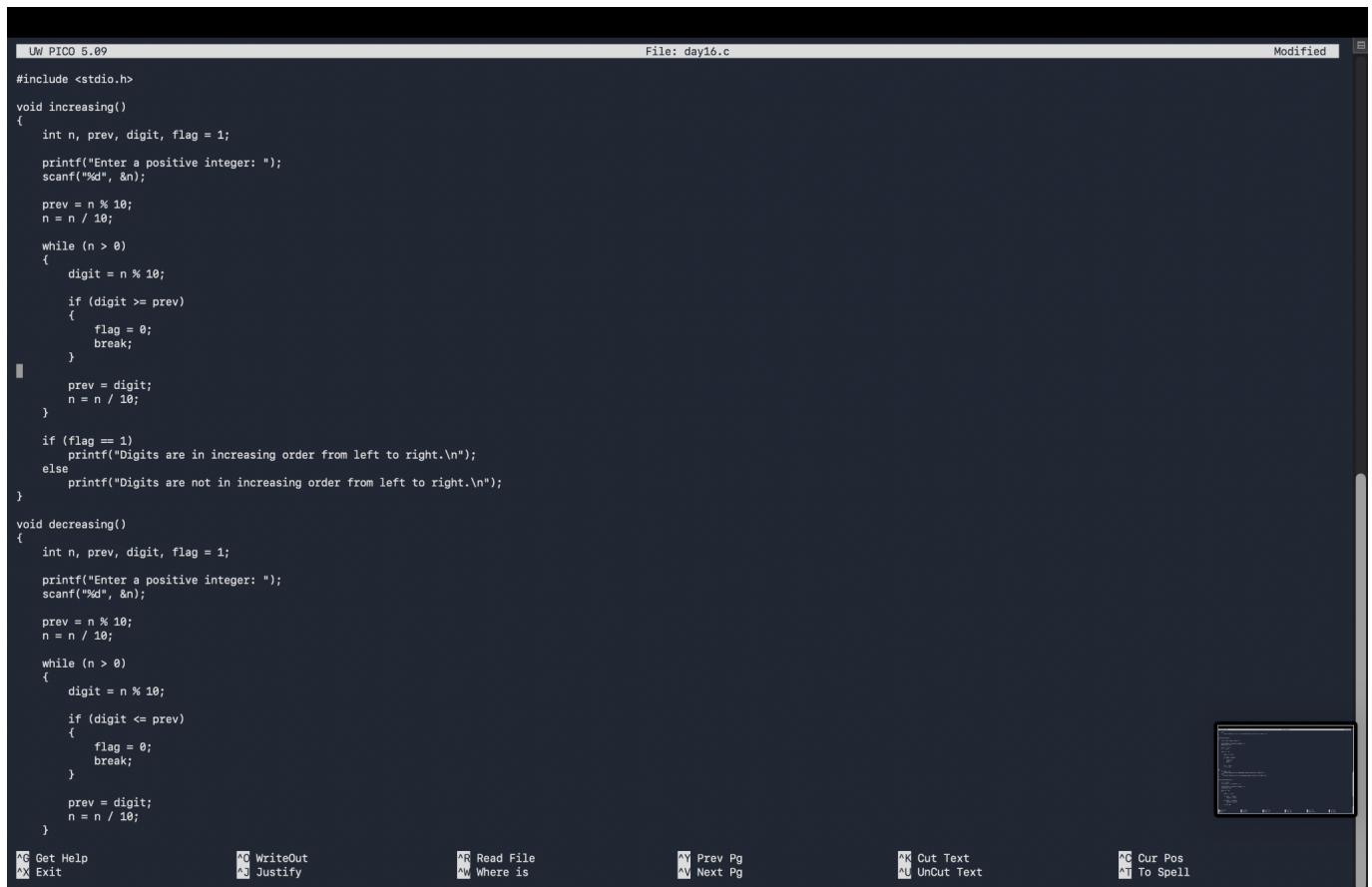
Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1 - Increasing	Choice: 1 Number: 1234	Digits are in increasing order	Digits are in increasing order	Pass
2 - Decreasing	Choice: 2 Number: 9871	Digits are in decreasing order	Digits are in decreasing order	Pass
3 - Largest/Smallest	Choice: 3 Number: 58320	Largest = 8 Smallest = 0	Largest = 8 Smallest = 0	Pass
4 - Second values	Choice: 4 Number: 58320	Second largest = 5 Second smallest = 2	Second largest = 5 Second smallest = 2	Pass

Program Screenshots

The following screenshots show the C source code written in the terminal editor and its execution.

Code / Execution Screenshot 1



The screenshot shows a terminal editor window titled "LW PICO 5.09" with the file "day16.c" open. The code defines two functions, "increasing()" and "decreasing()", which take an integer "n" and check if its digits are in increasing or decreasing order. The "increasing()" function uses a while loop to extract digits from right to left, comparing each digit to the previous one. If the digits are in increasing order, it prints "Digits are in increasing order from left to right.\n"; otherwise, it prints "Digits are not in increasing order from left to right.\n". The "decreasing()" function follows a similar logic but checks for decreasing order. The code is as follows:

```
#include <stdio.h>

void increasing()
{
    int n, prev, digit, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    prev = n % 10;
    n = n / 10;

    while (n > 0)
    {
        digit = n % 10;

        if (digit >= prev)
        {
            flag = 0;
            break;
        }

        prev = digit;
        n = n / 10;
    }

    if (flag == 1)
        printf("Digits are in increasing order from left to right.\n");
    else
        printf("Digits are not in increasing order from left to right.\n");
}

void decreasing()
{
    int n, prev, digit, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    prev = n % 10;
    n = n / 10;

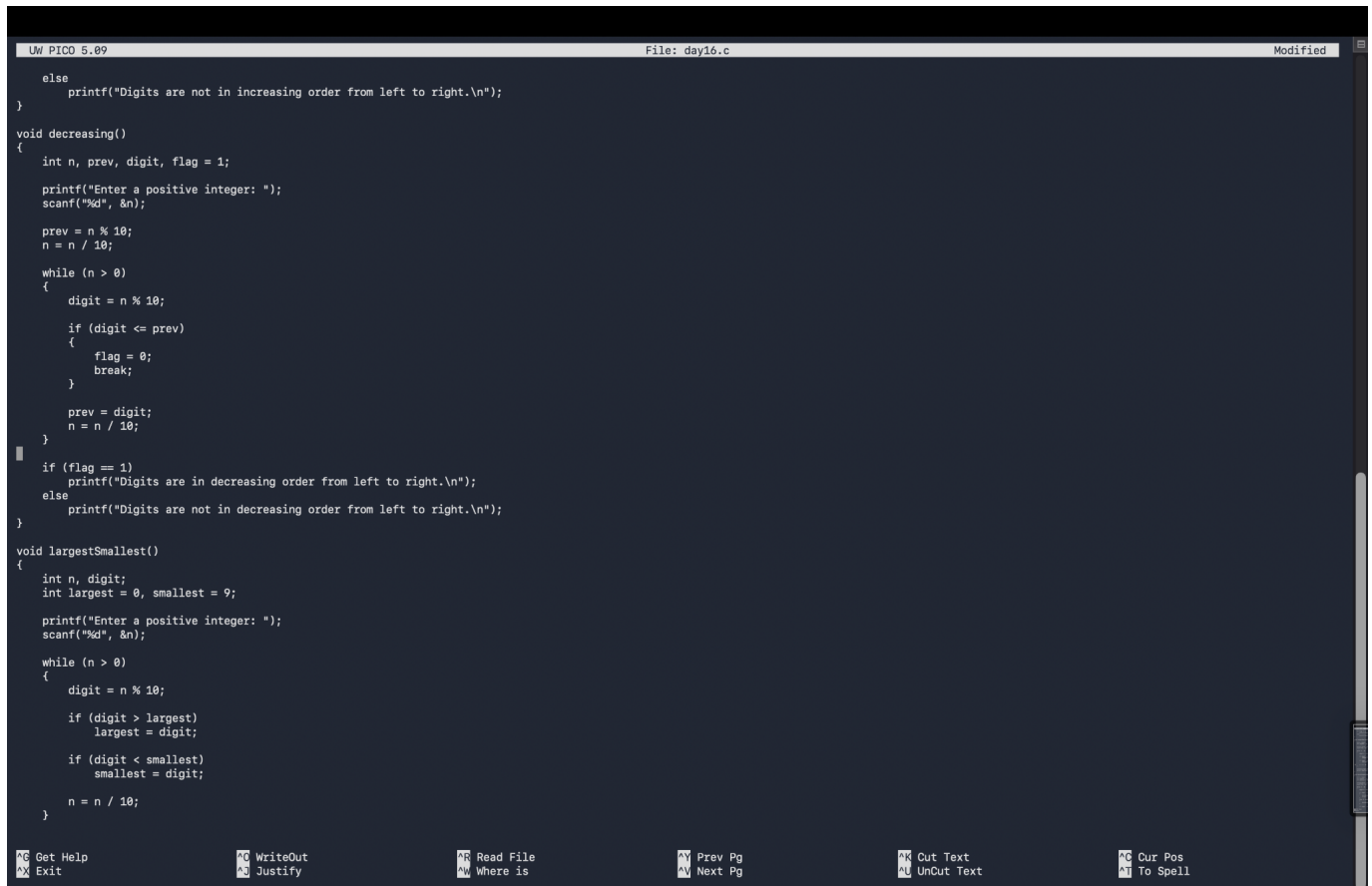
    while (n > 0)
    {
        digit = n % 10;

        if (digit <= prev)
        {
            flag = 0;
            break;
        }

        prev = digit;
        n = n / 10;
    }
}
```

The terminal editor has a status bar at the bottom with various icons and labels: "Get Help", "Exit", "WriteOut", "Justify", "Read File", "Where is", "Prev Pg", "Next Pg", "Cut Text", "UnCut Text", "Cur Pos", and "To Spell".

Code / Execution Screenshot 2



```
UW PICCO 5.09                                     File: day16.c                               Modified
}
    else
        printf("Digits are not in increasing order from left to right.\n");
}

void decreasing()
{
    int n, prev, digit, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    prev = n % 10;
    n = n / 10;

    while (n > 0)
    {
        digit = n % 10;

        if (digit <= prev)
        {
            flag = 0;
            break;
        }

        prev = digit;
        n = n / 10;
    }

    if (flag == 1)
        printf("Digits are in decreasing order from left to right.\n");
    else
        printf("Digits are not in decreasing order from left to right.\n");
}

void largestSmallest()
{
    int n, digit;
    int largest = 0, smallest = 9;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;

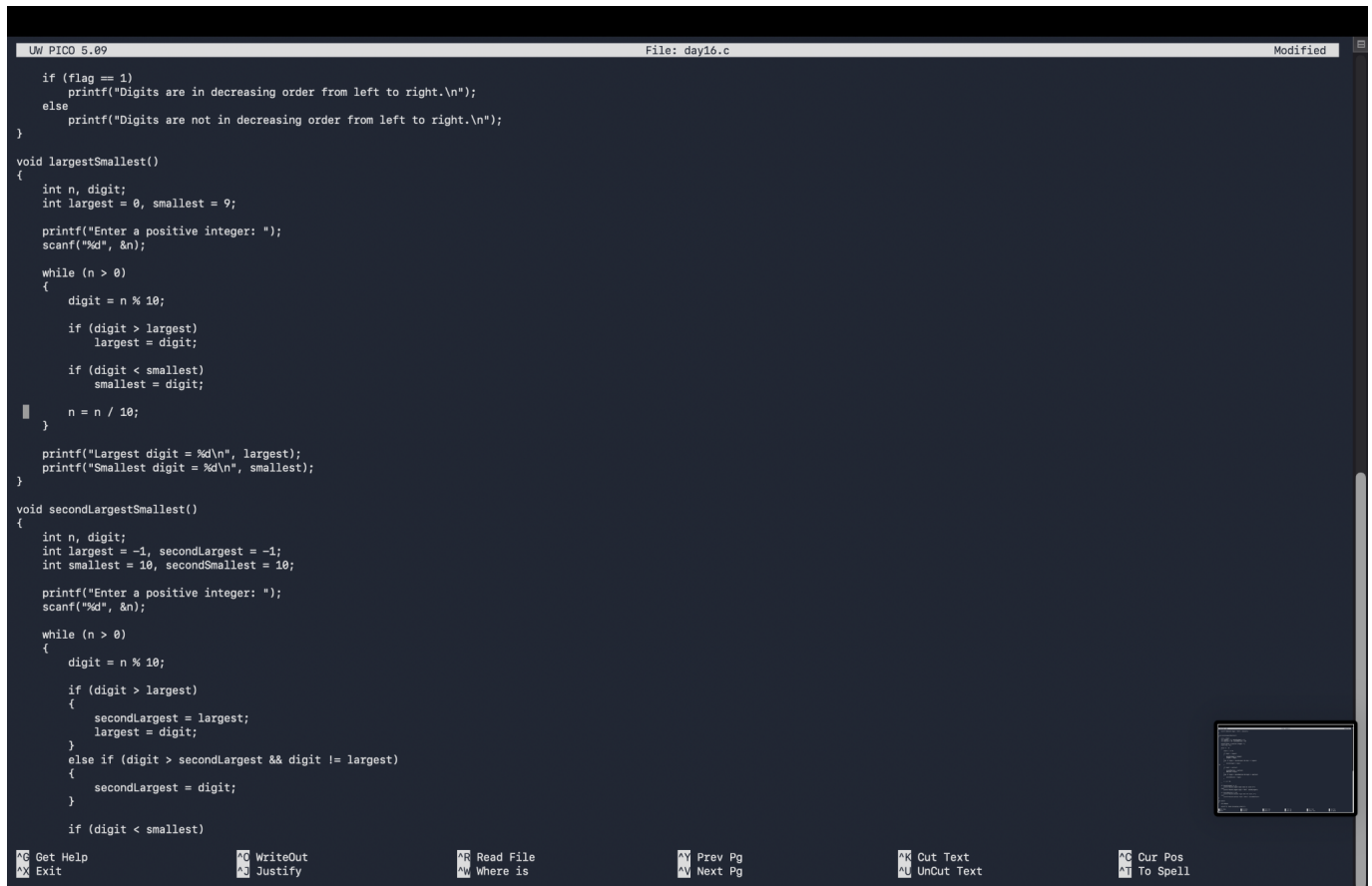
        if (digit > largest)
            largest = digit;

        if (digit < smallest)
            smallest = digit;

        n = n / 10;
    }
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

Code / Execution Screenshot 3



```
UW PICOD 5.09                                     File: day16.c                                     Modified
if (flag == 1)
    printf("Digits are in decreasing order from left to right.\n");
else
    printf("Digits are not in decreasing order from left to right.\n");
}

void largestSmallest()
{
    int n, digit;
    int largest = 0, smallest = 9;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;

        if (digit > largest)
            largest = digit;

        if (digit < smallest)
            smallest = digit;

        n = n / 10;
    }

    printf("Largest digit = %d\n", largest);
    printf("Smallest digit = %d\n", smallest);
}

void secondLargestSmallest()
{
    int n, digit;
    int largest = -1, secondLargest = -1;
    int smallest = 10, secondSmallest = 10;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;

        if (digit > largest)
        {
            secondLargest = largest;
            largest = digit;
        }
        else if (digit > secondLargest && digit != largest)
        {
            secondLargest = digit;
        }

        if (digit < smallest)
```

Get Help
Exit
WriteOut
Justify
Read File
Where is
Prev Pg
Next Pg
Cut Text
UnCut Text
Cur Pos
To Spell

Code / Execution Screenshot 4

```
UW PIC0 5.09 File: day16.c Modified
    printf("Smallest digit = %d\n", smallest);
}

void secondLargestSmallest()
{
    int n, digit;
    int largest = -1, secondLargest = -1;
    int smallest = 10, secondSmallest = 10;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;

        if (digit > largest)
        {
            secondLargest = largest;
            largest = digit;
        }
        else if (digit > secondLargest && digit != largest)
        {
            secondLargest = digit;
        }

        if (digit < smallest)
        {
            secondSmallest = smallest;
            smallest = digit;
        }
        else if (digit < secondSmallest && digit != smallest)
        {
            secondSmallest = digit;
        }

        n = n / 10;
    }

    if (secondLargest == -1)
        printf("Second largest digit does not exist.\n");
    else
        printf("Second largest digit = %d\n", secondLargest);

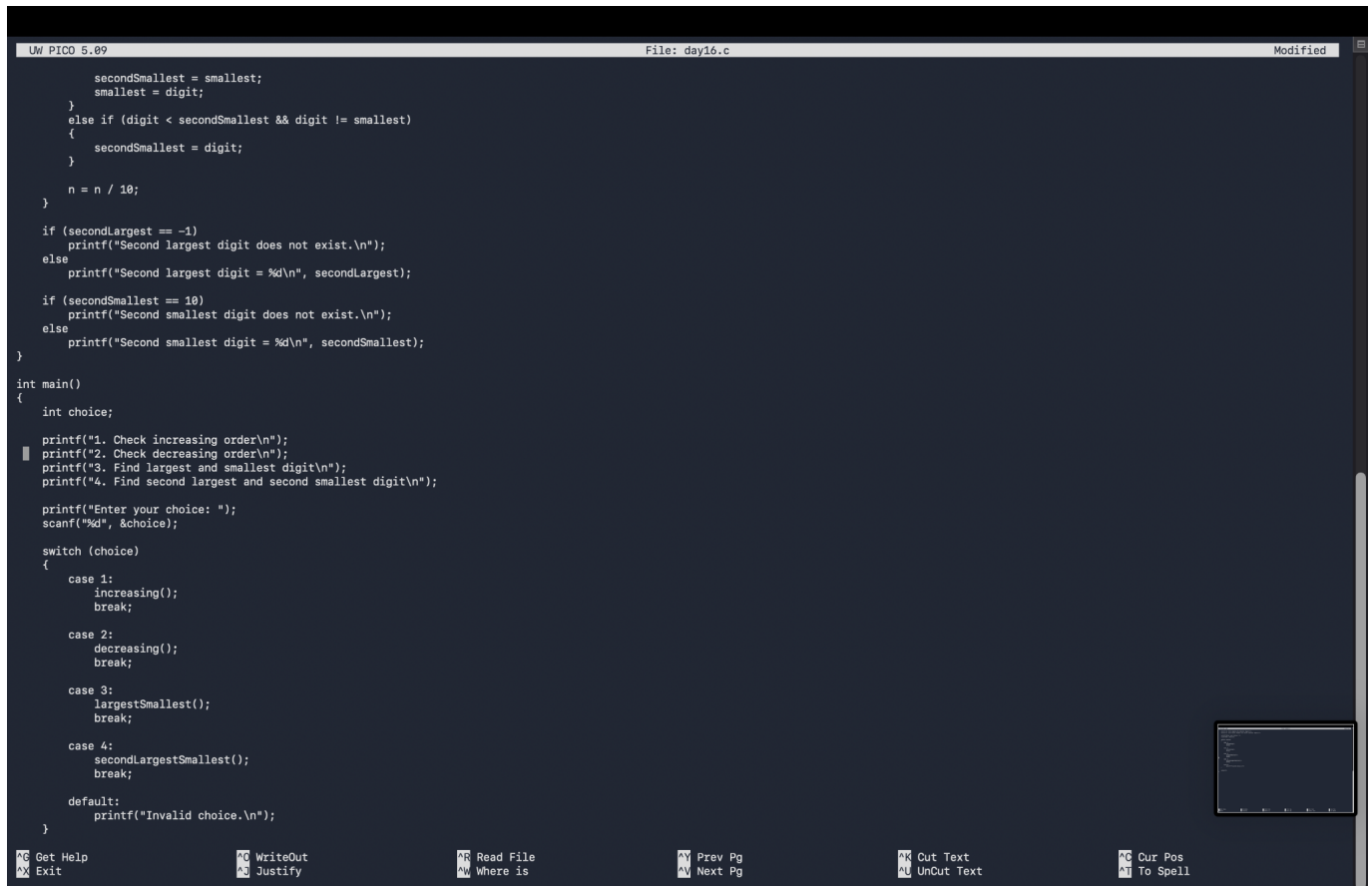
    if (secondSmallest == 10)
        printf("Second smallest digit does not exist.\n");
    else
        printf("Second smallest digit = %d\n", secondSmallest);
}

int main()
{
    int choice;

    printf("1. Check increasing order\n");

    Get Help    WriteOut    Read File    Prev Pg    Cut Text    Cur Pos
    Exit        Justify     Where is     Next Pg    UnCut Text  To Spell
```

Code / Execution Screenshot 5



```
UW PIC0 5.09 File: day16.c Modified
    secondSmallest = smallest;
    smallest = digit;
}
else if (digit < secondSmallest && digit != smallest)
{
    secondSmallest = digit;
}

n = n / 10;
}

if (secondLargest == -1)
    printf("Second largest digit does not exist.\n");
else
    printf("Second largest digit = %d\n", secondLargest);

if (secondSmallest == 10)
    printf("Second smallest digit does not exist.\n");
else
    printf("Second smallest digit = %d\n", secondSmallest);
}

int main()
{
    int choice;

    printf("1. Check increasing order\n");
    printf("2. Check decreasing order\n");
    printf("3. Find largest and smallest digit\n");
    printf("4. Find second largest and second smallest digit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            increasing();
            break;

        case 2:
            decreasing();
            break;

        case 3:
            largestSmallest();
            break;

        case 4:
            secondLargestSmallest();
            break;

        default:
            printf("Invalid choice.\n");
    }
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

Code / Execution Screenshot 6

```
UW PICD 5.09                                     File: day16.c                               Modified
printf("3. Find largest and smallest digit\n");
printf("4. Find second largest and second smallest digit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        increasing();
        break;

    case 2:
        decreasing();
        break;

    case 3:
        largestSmallest();
        break;

    case 4:
        secondLargestSmallest();
        break;

    default:
        printf("Invalid choice.\n");
}

return 0;
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

Code / Execution Screenshot 7

```
Last login: Thu Sep 10 17:16:50 on console
dhruvpandey@DHURVs-MacBook-Air ~ % nano day16.c
dhruvpandey@DHURVs-MacBook-Air ~ % clang day16.c -oday16
dhruvpandey@DHURVs-MacBook-Air ~ % ./day16
1. Check increasing order
2. Check decreasing order
3. Find largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 1
Enter a positive integer: 2
Digits are in increasing order from left to right.
dhruvpandey@DHURVs-MacBook-Air ~ %
```